

# PERMUTATIONS & COMBINATIONS

Complete Theory, Solved Examples & 25 Practice Questions

## 1. Essential Theory & Concepts

### Fundamental Principle of Counting

**Multiplication Rule:** If an operation can be performed in  $m$  different ways, and following which a second operation can be performed in  $n$  different ways, then the two operations in succession can be performed in  $m \times n$  different ways.

**Addition Rule:** If an operation can be performed in  $m$  ways and another operation can be performed in  $n$  ways, and they cannot be performed simultaneously, then either of the operations can be performed in  $m + n$  ways.

### Factorial Notation

The product of the first  $n$  natural numbers is denoted by  $n!$  (read as "n factorial").

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 3 \times 2 \times 1$$

**Note:**  $0! = 1$ .

### Permutations (Arrangements)

A permutation is an arrangement of a number of objects in a definite order. Order matters!

- **Formula:** The number of permutations of  $n$  different objects taken  $r$  at a time is given by:

$${}^n P_r = n! / (n - r)!$$

- **Permutation of objects when all are not distinct:** If there are  $n$  objects where  $p_1$  are of one kind,  $p_2$  of second kind, etc., then total arrangements =  $n! / (p_1! \times p_2! \times \dots)$ .
- **Circular Permutation:** Number of ways to arrange  $n$  distinct objects in a circle is  $(n - 1)!$ . (If clockwise and anti-clockwise are the same, e.g., a necklace, it is  $(n - 1)! / 2$ ).

## Combinations (Selections)

A combination is a selection of objects without regard to order. Order does NOT matter!

- **Formula:** The number of combinations of  $n$  different objects taken  $r$  at a time is:  

$${}^nC_r = n! / (r! \times (n - r)!)$$
- **Key Properties:**
  1.  ${}^nC_r = {}^nC_{n-r}$
  2.  ${}^nC_0 = {}^nC_n = 1$
  3.  ${}^nC_a = {}^nC_b$  implies  $a = b$  or  $a + b = n$ .

## 2. Solved Examples

Here are 12 detailed step-by-step examples demonstrating the application of these formulas.

### Example 1: Basic Evaluation

Evaluate  ${}^8P_3$  and  ${}^8C_3$ .

#### Solution:

For Permutation:  ${}^8P_3 = 8! / (8 - 3)! = 8! / 5! = (8 \times 7 \times 6 \times 5!) / 5! = 8 \times 7 \times 6 = 336$ .

For Combination:  ${}^8C_3 = 8! / (3! \times 5!) = (8 \times 7 \times 6) / (3 \times 2 \times 1) = 336 / 6 = 56$ .

### Example 2: Arranging Letters (No repetitions)

In how many ways can the letters of the word "PENCIL" be arranged?

#### Solution:

The word "PENCIL" has 6 distinct letters.

The number of arrangements is taking all 6 at a time.

Total ways =  $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$ .

### Example 3: Arranging Letters (With repetitions)

In how many ways can the letters of the word "SUCCESS" be arranged?

#### Solution:

Total letters,  $n = 7$ .

Repetitions: 'S' appears 3 times, 'C' appears 2 times, 'U' and 'E' appear 1 time each.

Total arrangements =  $7! / (3! \times 2!) = 5040 / (6 \times 2) = 5040 / 12 = 420$  ways.

### Example 4: Condition-based Arrangement

In how many ways can the letters of the word "MACHINE" be arranged so that the vowels always come together?

#### Solution:

Vowels in MACHINE: A, I, E (3 vowels). Consonants: M, C, H, N (4 consonants).

Group the vowels together as a single unit: (AIE).

Now we have 5 units to arrange: M, C, H, N, and (AIE). They can be arranged in  $5! = 120$  ways.

The 3 vowels can arrange among themselves in  $3! = 6$  ways.

Total ways =  $120 \times 6 = 720$ .

### Example 5: Number Formation

How many 3-digit numbers can be formed from the digits 1, 2, 3, 4, 5 if repetition of digits is not allowed?

#### Solution:

We need to arrange 3 digits out of 5 distinct digits. Order matters.

This is a permutation problem:  ${}^5P_3$ .

$${}^5P_3 = 5! / (5-3)! = 5! / 2! = (5 \times 4 \times 3 \times 2!) / 2! = 60 \text{ numbers.}$$

### Example 6: Circular Permutation

In how many ways can 6 people be seated around a circular table?

#### Solution:

For a circular arrangement of  $n$  distinct objects, the formula is  $(n - 1)!$ .

Here,  $n = 6$ .

Total ways =  $(6 - 1)! = 5! = 120$  ways.

### Example 7: Basic Selection (Committee)

In how many ways can a committee of 4 members be formed from a group of 10 people?

#### Solution:

Since we are only selecting (order doesn't matter), we use combinations.

We need to select 4 out of 10:  ${}^{10}C_4$ .

$${}^{10}C_4 = 10! / (4! \times 6!) = (10 \times 9 \times 8 \times 7) / (4 \times 3 \times 2 \times 1) = 210 \text{ ways.}$$

### Example 8: Selection with Categories

A box contains 5 red balls and 4 blue balls. In how many ways can 3 red and 2 blue balls be selected?

#### Solution:

Select 3 red out of 5:  ${}^5C_3 = 5! / (3! \times 2!) = 10$  ways.

Select 2 blue out of 4:  ${}^4C_2 = 4! / (2! \times 2!) = 6$  ways.

By the multiplication rule, total ways =  $10 \times 6 = 60$  ways.

### Example 9: Handshakes

There are 12 persons in a party. If every person shakes hands with every other person exactly once, what is the total number of handshakes?

#### Solution:

A handshake requires 2 people. The number of handshakes is the number of ways to choose 2 people out of 12.

${}^{12}C_2 = (12 \times 11) / (2 \times 1) = 66$  handshakes.

### Example 10: Diagonals of a Polygon

How many diagonals does an octagon (8 sides) have?

#### Solution:

To form a line segment, we select 2 points out of 8:  ${}^8C_2 = 28$  lines total.

Out of these 28 lines, 8 are the sides of the octagon. The rest are diagonals.

Number of diagonals =  $28 - 8 = 20$ .

(Shortcut Formula:  $n(n-3)/2 = 8(5)/2 = 20$ ).

### Example 11: At Least One Condition

In how many ways can a team of 3 be chosen from 4 boys and 3 girls so that the team contains at least 1 girl?

#### Solution:

Total ways to choose 3 without restriction:  ${}^7C_3 = 35$ .

Ways to choose a team with NO girls (i.e., all 3 are boys):  ${}^4C_3 = 4$ .

Ways to choose at least 1 girl = Total ways - Ways with no girls =  $35 - 4 = 31$  ways.

### Example 12: Finding 'n' or 'r'

If  ${}^nC_8 = {}^nC_2$ , find  $n$ .

#### Solution:

We use the property: If  ${}^nC_x = {}^nC_y$ , then either  $x = y$  or  $x + y = n$ .

Since  $8 \neq 2$ , it must be that  $8 + 2 = n$ .

Therefore,  $n = 10$ .

## 3. Practice Questions

Solve these 25 questions to master permutations and combinations. The answer key is provided on the last page.

**Q1. What is the value of  $0!$ ?**

- |                |       |
|----------------|-------|
| A) 0           | B) 1  |
| C) Not defined | D) 10 |

**Q2. Evaluate  ${}^6P_2$ .**

- |       |       |
|-------|-------|
| A) 15 | B) 30 |
| C) 12 | D) 36 |

**Q3. Evaluate  ${}^7C_3$ .**

- |       |        |
|-------|--------|
| A) 35 | B) 210 |
| C) 21 | D) 42  |

**Q4. If  ${}^nC_4 = {}^nC_6$ , what is n?**

- |       |       |
|-------|-------|
| A) 4  | B) 6  |
| C) 10 | D) 24 |

**Q5. In how many ways can 5 boys be seated in a row?**

- |        |       |
|--------|-------|
| A) 24  | B) 60 |
| C) 120 | D) 25 |

**Q6. How many words can be formed using all letters of the word 'APPLE'?**

- |        |       |
|--------|-------|
| A) 120 | B) 60 |
| C) 24  | D) 30 |

**Q7. In how many ways can a committee of 3 be chosen from 8 people?**

- |       |        |
|-------|--------|
| A) 56 | B) 336 |
| C) 24 | D) 112 |

**Q8. How many 3-digit numbers can be formed using 1, 2, 3, 4 without repetition?**

- |       |       |
|-------|-------|
| A) 12 | B) 24 |
| C) 64 | D) 48 |

**Q9. In how many ways can 4 people sit around a circular table?**

- |       |       |
|-------|-------|
| A) 24 | B) 12 |
| C) 6  | D) 4  |

**Q10. How many handshakes are possible if there are 10 people in a room and everyone shakes hands once?**

- |        |       |
|--------|-------|
| A) 90  | B) 45 |
| C) 100 | D) 20 |

**Q11. What is the value of  $^{100}C_{100}$ ?**

- |        |       |
|--------|-------|
| A) 100 | B) 0  |
| C) 1   | D) 99 |

**Q12. How many diagonals does a hexagon have?**

- |       |       |
|-------|-------|
| A) 6  | B) 9  |
| C) 12 | D) 15 |

**Q13. A bag has 3 red and 4 black balls. Ways to select 2 red and 1 black ball?**

- |       |       |
|-------|-------|
| A) 12 | B) 7  |
| C) 35 | D) 14 |

**Q14. Number of ways to arrange the letters of 'MATHS'?**

- |        |        |
|--------|--------|
| A) 25  | B) 60  |
| C) 120 | D) 720 |

**Q15. How many ways to select a captain and a vice-captain from a team of 11?**

- |        |       |
|--------|-------|
| A) 110 | B) 55 |
| C) 121 | D) 22 |

**Q16. If  ${}^nP_2 = 20$ , find n.**

- |      |      |
|------|------|
| A) 4 | B) 5 |
| C) 6 | D) 8 |

**Q17. Ways to arrange 3 books out of 7 on a shelf?**

- |        |        |
|--------|--------|
| A) 35  | B) 210 |
| C) 343 | D) 42  |

**Q18. From a deck of 52 cards, in how many ways can 2 cards be drawn?**

- |         |         |
|---------|---------|
| A) 1326 | B) 2652 |
| C) 104  | D) 52   |

**Q19. How many 4-digit PINs can be formed if digits can repeat? (0-9)**

- |         |          |
|---------|----------|
| A) 5040 | B) 10000 |
| C) 9000 | D) 9999  |

**Q20. In how many ways can 2 identical red pens and 3 identical blue pens be arranged?**

- |        |       |
|--------|-------|
| A) 10  | B) 20 |
| C) 120 | D) 5  |

**Q21. A question paper has 2 sections of 5 questions each. Select 2 from each section. How many ways?**

- |        |        |
|--------|--------|
| A) 20  | B) 50  |
| C) 100 | D) 200 |

**Q22. How many ways can 4 letters be posted in 3 post boxes?**

- |       |       |
|-------|-------|
| A) 12 | B) 64 |
| C) 81 | D) 7  |

**Q23. Arrangements of word 'DOG' where 'O' is always at the center?**

- |      |      |
|------|------|
| A) 1 | B) 2 |
| C) 3 | D) 6 |

**Q24. Number of triangles that can be formed by joining 6 non-collinear points?**

- |       |       |
|-------|-------|
| A) 15 | B) 20 |
| C) 30 | D) 12 |

**Q25. Value of  ${}^5P_5$  is?**

- |        |      |
|--------|------|
| A) 1   | B) 5 |
| C) 120 | D) 0 |

## Answer Key

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**Q1: B**

**Q2: B**

**Q3: A**

**Q4: C**

**Q5: C**

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**Q6: B**

**Q7: A**

**Q8: B**

**Q9: C**

**Q10: B**

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**Q11: C**

**Q12: B**

**Q13: A**

**Q14: C**

**Q15: A**

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**Q16: B**

**Q17: B**

**Q18: A**

**Q19: B**

**Q20: A**

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**Q21: C**

**Q22: C**

**Q23: B**

**Q24: B**

**Q25: C**

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